Octo

An OctoStudio companion book



20 projects. Your drawings. Your voice. Your code.

Circuit Kids

Welcome to Ingwane

Every platform and device in Circuit Kids should have a companion book. This book turns that principle into a beginner pathway: first find your way around the app, then make something that matters.

About this edition

1. Target readers: ages 7-12, with a partner or adult when needed.
Younger children can join in through adult-led drawing, storytelling and play.

2. Story layer: Thandi, Sipho, Lerato and Gogo visit uShaka Marine World in a fictional story arc. Project 1 has octopuses; later projects bring in other marine life and care for the sea.

3. Structure: a navigation section, then 20 two-page projects: story and making, then code and testing. Project 20 includes an extra testing and reflection page.

4. Page format: A4 portrait, spacious reading text and schematic code recipes.
Read on a phone or tablet, or print for learning together.

5. Title and license: Ingwane. The book is licensed under CC BY-NC-SA 4.0; see the copyright and license notice below.

What this book is, and is not

App facts were checked against the July 2026 official reference guide for OctoStudio 1.1.0 and higher. These original projects have not been run on physical devices or with children. Expected results are test targets, not claims that testing has passed.

Before physical printing: device trials, learner and language checks, rights checks and print setup.
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Your journey

Read the story first. Let it stay with you. Then make one piece of that world move, speak or respond.

Find your feet

Before Project 1: get the app ready, meet the screens, make your first stack, learn to recover a mistake, and keep a safe copy of your work.

20 projects

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02 The taxi waits / Motion and order / p. 11
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05 Gogo's hidden eel / Hide, show and two sprites / p. 17
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A grown-up helps us begin

Ready before children arrive

1. Install OctoStudio from the official app-store links at octostudio.org using internet access. Update before class; costumes need version 1.1.0 or later. Open the app once and check the chosen language.

2. Use a supported phone or tablet: the current guide lists Android 8+, iOS 15+, or Chromebooks with Android apps. Other ordinary laptops and desktops are not supported by this app. Check actual school devices before promising a course.

3. After installation, creating works offline. Keep space available for photos, audio and project files. Sending a file online may need data; offline creation is not the same as free online sharing.

4. Pair learners: one is the maker touching the screen, one is the guide reading or pointing. Swap each project. Below age 7, use adult-led drawing, storytelling and one-block play rather than expecting the whole sequence independently.

5. In Home > Settings, turn on Starter Cues for beginners. Show Screen Taps helps demonstrations. Try device VoiceOver or TalkBack and the app's code-block sound effects with the learner who needs them. Test access rather than assume one setting suits everyone.

Our safety promise

Use drawings and library sounds first. Ask before taking someone's photograph or recording their voice. Do not include names of schools, addresses, phone numbers or private conversations in projects. An adult reviews anything leaving the device. Never use this app while crossing a road. Hold devices securely over a soft surface for tilt or shake. No vigorous shaking.

The app's privacy policy says data is stored locally and not collected by OctoStudio. A file exported to another service is a separate disclosure. Back up before uninstalling; uninstalling can delete local app data. No public sharing is needed to complete this book.

Get the free OctoStudio app

Scan the code for your device with the camera, or type the web address underneath. The app is made by the MIT Lifelong Kindergarten group (the Scratch team) and links come from octostudio.org.

Android (Google Play)

https://play.google.com/store/apps/details?id=edu.mit.octostudio

iPhone and iPad (App Store)

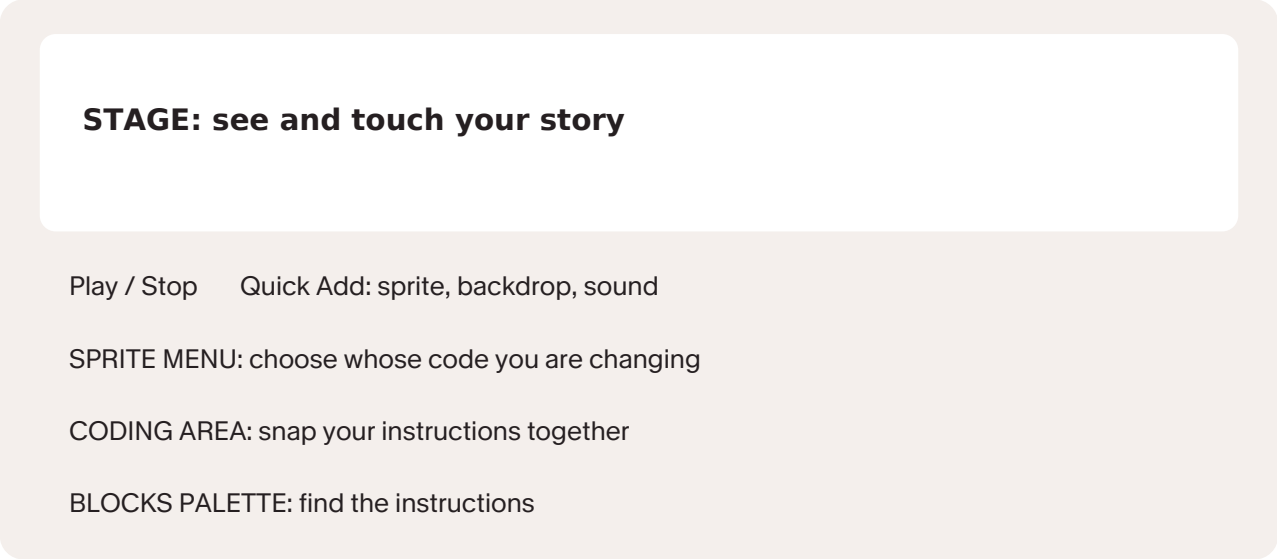
https://apps.apple.com/app/octostudio/id6461573316

Meet the two screens

Home: your shelf of projects

Home lists your projects. Create a new project starts a fresh one. Explore has examples to play with. The gear opens Settings. A project's menu offers Duplicate, Favorite, Share and Delete. Try Duplicate on a practice project, not Delete on your best work.

Editor: your little theatre



Original simplified map, not a screenshot. On Chromebooks and larger tablets the same parts can be arranged side by side.

A sprite is a character or object you can code. A backdrop is the scene behind it. The stage is where the project runs. A block is one instruction. A stack, or script, is a connected set of blocks for one sprite.

Tap a sprite in the sprite menu before coding it. The stage shows the whole story, but the coding area belongs to the selected sprite. If the wrong character moves, check the selection first.

South African languages in OctoStudio

OctoStudio's current language list includes Afrikaans, English, isiXhosa, isiZulu, Sepedi (Northern Sotho), Setswana (Tswana) and Tshivenḓa (Venda). Choose the app language in Home > Settings. Learners can tell their stories in the languages they use at home; ask a speaker to check greetings and recordings.

Source: octostudio.org/about | Language list checked 10 October 2026.

Make your first instruction

1. Create a new project from Home. Choose a backdrop from the library. Use Add sprite and choose an emoji or Paint a shape. Keep just one sprite for this practice.

2. Select your sprite in the sprite menu. Find the "When to start?" category in the palette. Drag the Play-start block into the coding area.

3. Open Motion. Drag a Jump block below the start block until the pieces snap together. Tap its number and try 20.

4. Press Play under the stage. Your sprite should jump. Press Stop. Now change 20 to 40 and press Play again. What changed?

5. Drag a second Jump block below the first. Predict what two jumps will look like. Play and compare. If nothing happens, reconnect the blocks and check the selected sprite.

READ DOWN THE STACK

When Play starts

Jump 20

Jump 40

Code is a plan the device can follow. The start block answers "When?" The next blocks answer "Do what?" Changing one number changes the plan. Testing means trying the plan and watching closely.

Recipes in this book use descriptive words for blocks and their menu choices. They are not literal typed code, app screenshots or a promise that every language uses the same labels. Use the official guide to match the block icons. Indented lines go inside a loop or condition's mouth.

Your block toolbox

When to start?

Play, tap, shake, touching a sprite, a backdrop change. An event tells a stack when to begin.

Motion

Move, jump, turn, go to a stage position, face a direction. Use a small move before a big one.

Words and Sounds

Speech bubble, stage text, voice aloud, recorded or library sound. Play sound "until done" when the next action must wait.

Scene

Size, costumes, hide/show, backdrop, front/behind. Costumes are different looks of the same sprite.

Colors and Light

Glow, colour changes, flashlight. This book uses on-screen light so a torch is not required.

Control

Repeat, forever, if, wait, stop, send and receive a message. These blocks organise the order.

More Blocks

Tilt, variables, numbers, beam and custom blocks. Add complexity only when your story needs it.

Some blocks have slow, medium, fast and instant speed options. "Set size" chooses a total size; "change size" adds or subtracts from the current size. "Set score" replaces the number; "change score" adds to it. These differences matter when restarting.

Keep your work. Fix your mistakes.

A project is more than a video

1. Finish a small change, press Stop, then use Home. Find your project in the Home list and reopen it. Check that your latest work is still there on this device.

2. Before a big experiment, use the project menu > Duplicate. Open the copy and change it. Keep the original safe.

3. For a backup, open Share > Send Project File. Choose the device's local file-saving option if available. Save a copy in a folder an adult can find. The guide confirms saving a project file to the device; exact share-sheet buttons vary.

4. Test the backup: open that saved .octostudio file with OctoStudio and check its sprites and blocks. Receiving a project file needs OctoStudio. A video or GIF is only a recording, not editable code.

When something goes wrong

Nothing runs? Check for a start block, a snapped connection, and the correct trigger. Wrong sprite? Select it in the sprite menu. Lost blocks off-screen? Zoom out or tidy the coding area. Wrong edit? Try Undo, then Redo if needed. Missing sprite? Check Hide and size. Silent audio? Check sound selection, volume and permissions. If an effect changes after every run, reset its starting value in the Play stack.

Stop before moving a sprite back to its start. Press Play again for a fresh test. Do not delete a project to fix a small mistake. Tell your partner: "I expected ___. I saw ___. I will change ___."

Practice backup without personal photos or voices. Share locally in the room only if the adult agrees. Keeping a file safe does not require posting it online.

Sawubona Ngwane

Tap events and speech bubbles

Thandi had moved into the street only yesterday. At the gate she held a folded drawing and waited. Everyone seemed to know everyone. She knew nobody. Sipho spotted a purple shape peeping from the paper. "Is that a hat?" Thandi shook her head and opened it. An octopus spread eight wonky arms across the page. Sipho drew a smaller octopus beside it. "Sawubona Ngwane," Thandi whispered to her new paper friend. Sipho answered with a wave. By the time Gogo brought out water, the two octopuses had names and the children had a game. They imagined a sea where there was room for a newcomer. Thandi tucked the drawing into her pocket. One small hello had made room for a friend.

Keep the story with you

What could your character say to make someone feel welcome?

Make the scene

1. Create a new project and choose a quiet library backdrop. Add an octopus emoji sprite if available, or paint a friendly octopus.

2. Select your sprite in the sprite menu. Drag a Tap-start block into its coding area. Choose the option that means tapping this sprite, not anywhere.

3. In Words and Sounds, attach a speech-bubble block. Enter 'Sawubona Ngwane' and set its time to 2 seconds.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

YOUR GREETING OCTOPUS

When this sprite is tapped

Say in bubble: Sawubona Ngwane for 2 seconds

Why this works

The tap is an event. It starts this stack only when the selected sprite is tapped. The bubble is an action. Nothing needs to run before the tap.

Try, watch, fix

Press Play if needed to enter the running view. Tap the sprite, then an empty part of the backdrop. Only the sprite should give the greeting. Check the tap option if both respond.

Make it yours

Record greetings on paper in languages your family uses. Choose one for your sprite. Ask a speaker to check the words; do not rely on automatic voice pronunciation.

Tell someone

Show your partner how to select the sprite and find its code. Save a project-file copy after the first successful greeting.

The taxi waits

Motion, positions and sequence

At the end of the street, the taxi waited while Auntie tied a loose shoelace. Sipho wanted to race ahead. Today Thandi, Sipho, Lerato and Gogo were visiting uShaka Marine World in Durban. The driver lifted one hand. There was time to wait. At home later, Sipho drew the trip on paper, from the dusty stop to uShaka Marine World. Thandi placed their paper octopus beside the taxi window: a pretend passenger joining their visit as a paper friend. Sipho pushed the toy taxi too far and it fell off the page. Thandi marked a shorter distance. This time it stopped exactly at their drawn entrance. Gogo reminded them that real sea animals were not toys to collect and carry home. Their octopus belonged on paper. A good journey was more than going fast. It was knowing where to begin and where to stop.

Keep the story with you

Where should your story taxi start and finish?

Make the scene

1. Add a taxi sprite or draw a simple rectangle with wheels. Use a library backdrop. This is a screen story, not a road-crossing activity.

2. Select the taxi. Attach a Go-to-position block below Play. Tap its map pin and place the taxi near the left side of the stage, away from the edge.

3. Add Move right 60 at medium speed, then a speech bubble 'We have arrived!' for 2 seconds. Keep enough space for the move.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

TAXI

When Play starts

Go to your chosen left-side start (instant)

Move right 60 (medium)

Say in bubble: We have arrived! for 2 seconds

Why this works

Read down: first reset the taxi, then move, then speak. The Go-to block chooses a position. Move changes that position by a distance. The reset makes every run begin in the same place.

Try, watch, fix

Play twice. The taxi should start at the same place each time. Change 60 to 30 and compare. If it wraps at an edge, move the start inward or shorten the distance.

Make it yours

Draw the journey on paper. Estimate a distance, try it, then adjust. Change the backdrop to a place from your own story, without photographing an address.

Tell someone

Explain which block you would move if the taxi spoke before arriving.

A sea star in the wind

Paint tools and counted repetition

After their visit to the aquarium at uShaka Marine World, Lerato made a paper sea star for the street's puppet show. She painted it orange and spread its arms across the page. Then the wind came. The puppet turned once, twice, and landed beside Gogo's chair. Lerato chased it, laughing. She wanted to remember its dance without losing it again. On her next page she drew four little arrows. Each arrow turned a quarter of a circle. Together they brought the sea star back to the way it had begun. The wind had gone by lunchtime, but Lerato kept the arrows. At the puppet show she turned the sea star slowly with her fingers. Children who had missed the windy morning could still see the whole dance. Their sea story had found a way to move.

Keep the story with you

How many quarter-turns make a full turn?

Make the scene

1. Add sprite > Paint. Draw your own sea star puppet with a shape or brush. Use Fill if you want a solid colour. Finish the drawing so it appears in the sprite menu.

2. Select the sea star puppet. Below Play add a Repeat block from Control. Enter 4.

3. Put Turn clockwise 90 inside Repeat's mouth. Choose medium speed so you can see each turn.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

PAINTED SEA STAR

When Play starts

Repeat 4 times:

    Turn clockwise 90 (medium)

Why this works

Repeat runs the blocks inside it a counted number of times. Four turns of 90 degrees make 360 degrees. The turn block belongs inside the mouth, not below the outside of Repeat.

Try, watch, fix

Predict where the sea star puppet will face after one turn. Play with Repeat 1, then 2, then 4. The four-turn version should finish facing the way it began.

Make it yours

Paint a different pattern on your sea star. Try Repeat 6 with turns of 60 degrees. Draw the pattern before running it.

Tell someone

Let your partner point to the repeated instruction. A loop saves you from dragging the same turn block four times.

The tiny turtle

Size, wait and initial values

At uShaka Marine World, Thandi had watched the turtles. Back home, she painted a tiny turtle in the middle of a very large sea. Sipho leaned closer. "I can hardly see our friend." Thandi wanted it enormous at once, but Gogo suggested four smaller changes. Sipho marked each new size on a strip of paper. The flippers stretched a little farther across the scene each time. Soon the little turtle was easy to see without covering its whole home. Thandi set the drawing back to its smallest size and tried again. This time she knew what would happen. It was a pretend growing story, not a clock for real animals. Yet the small changes had taught her something useful: begin where you mean to begin, and give your reader time to notice.

Keep the story with you

How can a few small changes tell a growing story?

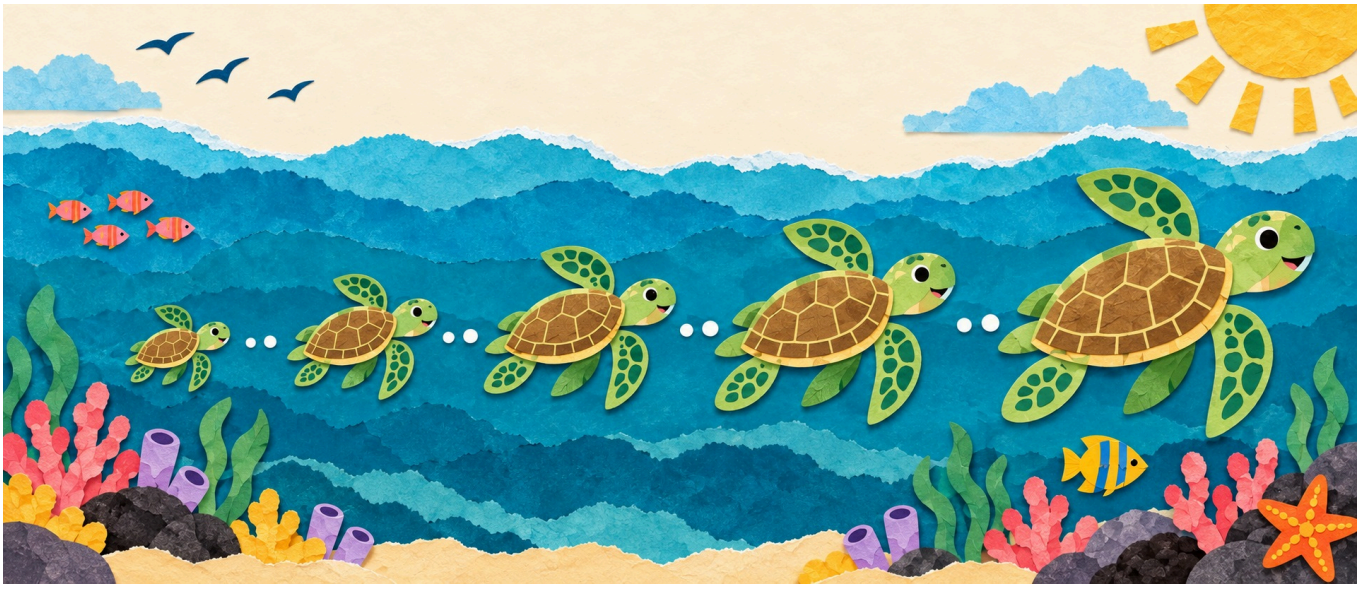
Make the scene

1. Paint a turtle sprite, or choose a turtle emoji if available. Select it.

2. Below Play add Set size 30 percent. Add Repeat 4.

3. Inside Repeat, put Wait 1 second followed by Change size by 10. Use a visible, medium speed.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

TURTLE

```
When Play starts
Set size to 30% (instant)
Repeat 4 times:
    Wait 1 second
    Change size by 10
```

Why this works

Set size gives a starting value. Change size adds to it. Starting at 30 and adding 10 four times gives 70 percent. Wait lets the viewer notice each step. This is a story animation, not a measurement of real turtle growth.

Try, watch, fix

Play twice and watch the size. It should begin small on both runs. Remove the reset only in a duplicate project and see why repeated runs can become confusing.

Make it yours

Make the turtle shrink back by changing the amount to -10 in a second loop. Keep the starting size large enough to stay visible.

Tell someone

Make a paper number line for 30, 40, 50, 60, 70. Tell your partner what the device is counting.

Gogo's hidden eel

Hide/show and coding the correct sprite

After the aquarium visit, Gogo made an eel puppet for the table-top sea. It vanished behind a tin on the table. Lerato searched the windowsill and the shelf. Nothing. Sipho moved the tin to make room for their painted sea, and its purple tail appeared. Lerato pulled out the puppet and brushed off a little flour. They placed it beside a paper rock, then tucked it behind the rock again. This time Sipho knew where to look. The children made a guessing game for their sea story: tap the rock and discover the friend behind it. Gogo watched them laugh at the same surprise twice. They did not need to lift real rocks or disturb an animal's home. A paper hiding place was enough to make a reader curious.

Keep the story with you

What should appear when your reader discovers a clue?

Make the scene

1. Add two sprites: an eel you paint and a painted rock or other clue. Keep both on the stage.

2. Select the eel sprite. Build its Play > Hide stack using instant speed. Add a separate Tap-start stack on this same sprite, but set its tap target to the rock sprite.

3. Under that tap event add Show at 100 percent and a bubble 'Found it!' for 2 seconds. The eel owns the actions even though the rock receives the tap.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

EEL: INITIAL SETUP

When Play starts

Hide (instant)

EEL: REVEAL

When painted rock is tapped

Show at 100% (instant)

Say in bubble: Found it! for 2 seconds

Why this works

One sprite can have more than one stack. The eel hides at Play and shows after the rock is tapped. A hidden sprite cannot interact, so do not require the hidden eel itself to receive the tap.

Try, watch, fix

Play, wait for the eel to hide, then tap the rock. The eel should appear. Stop and Play again: it should hide again. If the rock disappears, you coded the wrong sprite.

Make it yours

Draw a clue that is meaningful to your family. Make a second clue reveal another painted object, one action at a time.

Tell someone

Explain why hiding the only tappable clue would make the puzzle impossible.

A voice for the sea

Recording, trimming and audio order

The turtle exhibit at uShaka Marine World stayed in Thandi's thoughts. At home she drew a turtle beside a painted beach, but the drawing stayed quiet. She wanted it to speak for the sea. Gogo asked what it might say. Thandi tried a grand speech and lost her place. Then she tried one small sentence: "Keep our sea free of litter." The first recording had a long silence at the start. The second had a cough at the end. She kept the middle, where the words were clear. When Sipho tapped the turtle and heard Thandi's voice, he knew exactly what she wanted. The picture held their imaginary friend. The sound held her promise. Thandi had learned that her own voice was enough to tell a story she cared about.

Keep the story with you

What short sound could help someone imagine your scene?

Make the scene

1. Paint a turtle. Use Quick Add > Add sound. Choose Record and, with adult permission, record your own 3-5 second sea-care message. Do not record someone else without asking.

2. In the sound editor, drag the selection bars around the part you want to keep and use Crop. Listen. Re-record if needed. If microphone access is unavailable, choose a library sound.

3. Select your sprite. Attach Play sound beneath its Tap-start block. Select your recording and choose 'until done'. Add a bubble after it.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

TURTLE

When this sprite is tapped

Play your chosen sound (until done)

Say in bubble: Keep our sea free of litter! for 2 seconds

Why this works

'Until done' waits for the audio to finish before the next block. 'And continue' lets the next block begin while audio is playing. Recording is a creative choice; no personal voice is required.

Try, watch, fix

Tap once. The bubble should come after the recording finishes. Try 'and continue' in a duplicate and compare. Check volume and sound selection if silent.

Make it yours

Make a safe sound with paper or hands. Try Echo on a selected part, then compare with the original. Do not shout into the microphone.

Tell someone

Let your partner describe what the sound adds that the drawing alone cannot. Keep the file on the device unless an adult approves sharing.

The bottle-top band

Several sprites and independent events

Sipho found three clean bottle tops in the craft box. Before they became music, he asked Gogo where they should go afterwards. They agreed to keep them in the box, not leave them outside. Lerato drew a different sea animal on each cardboard drum pad. Blue made a soft click, red a sharper click, green a rustle. Soon the children were playing click, click, pause. When everyone played at once, the pattern disappeared into noise. They stopped, listened and gave each sea animal a turn. Their little band became part of a sea-care puppet show. Nothing needed to be thrown into water to make a splash in the story.

Keep the story with you

Can your reader perform a rhythm, not only watch one?

Make the scene

1. Paint three sea-animal sprites: a blue fish, a red crab and a green turtle. Use the sound library to add three short sounds, or record safe taps with permission.

2. Select each sea animal in turn. Give each its own Tap-start stack and a Play sound block with that animal's chosen sound.

3. Use 'until done' within each stack. Arrange the animals so fingers can tap them without overlapping.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

BLUE FISH

When this sprite is tapped

Play sound A (until done)

RED CRAB AND GREEN TURTLE

Make the same stack on each sprite

Choose a different sound for each

Why this works

Each sprite owns its code. Waiting in the blue stack does not automatically stop the red stack. Separate events can overlap. A clean rhythm comes from the player's timing as well as the code.

Try, watch, fix

Tap blue, wait, tap red. Then try blue and red quickly. Listen for overlap. If every animal sounds the same, check each sprite's sound menu.

Make it yours

Draw a rhythm strip: blue, blue, pause, green. Ask your partner to play it. Then invent a different pattern.

Tell someone

Offer a visual rhythm card for someone who cannot hear the sounds. Add a short glow to each tap stack so a sound also has a visible response.

A paper fish up close

Camera, photo selection and cropping

At uShaka Marine World's aquarium, Lerato had watched fish glide past the viewing glass. Back home she painted a fish with a fan-shaped tail for the puppet show. Beside it was a tiny drawn sign: "Our sea is home." She wanted to put the paper friend on her screen, so she laid it on plain paper and asked an adult to help take a photograph. In the picture, the fish looked small because the table took up so much room. She traced around its edge and removed the empty part. Now the little sign could be read. Thandi made a new drawing directly on the screen instead. Both fish belonged in their story. One carried a paper artwork into the device. One began there. Neither needed a real animal to be caught, touched or moved.

Keep the story with you

What detail of an ordinary object deserves a closer look?

Make the scene

1. With permission, Add sprite > Camera and photograph a paper fish you made on plain paper. Avoid faces, house numbers and private items. Photos lets you choose an existing approved image.

2. Trace and crop around the object, or use a simple crop shape. Use erase for unwanted parts. A painted fish is an equal alternative if camera access is denied.

3. Select the fish. Make a tap stack with Turn 20 clockwise, then a bubble describing one observed detail.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

FISH

When this sprite is tapped

Turn clockwise 20 (medium)

Say in bubble: Look at my fan-shaped tail! for 3 seconds

Why this works

A photo can become a sprite, not only a backdrop. Cropping changes which part of the photo you use. It does not make private information safe if it remains visible in the image.

Try, watch, fix

Tap once and check that the fish turns and speaks. Tap again: another turn adds to the first. This is deliberate. Stop and move it back if you want the original arrangement.

Make it yours

Paint an imagined fish next to the photo. Let each describe a different detail. Talk about observation and invention.

Tell someone

Before any export, inspect the whole image and listen to every sound for personal information. An adult makes the final sharing decision.

The turtle that waved

Costumes and frame animation

A paper turtle hung beside Thandi's window, but its flippers never moved. Sipho made another drawing with two flippers curled upward. Lerato placed the pictures beside each other. Open, curled. Open, curled. When they flicked between them, their turtle seemed to wave. One drawing had drifted higher than the other, so the whole friend jumped around. They put a small dot where its body should stay and drew again. This time the flippers changed while the body held its place. Thandi watched their paper sea friend long after the game was over. It was a made-up wave, not a science film of turtle movement. They had learned how carefully placed pictures could carry a greeting.

Keep the story with you

Which part should change, and which part should stay still?

Make the scene

1. Use OctoStudio 1.1.0 or higher. Paint a turtle sprite. Edit it from the sprite menu to open its costumes in the Paint Editor.

2. Duplicate the first costume. Change only the flipper position in the second. Onion skin can show the previous drawing so the body stays aligned.

3. Select the turtle. Below Play, add Repeat 4 with Play costumes inside. Set it to play your two visible costumes at medium speed.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

TURTLE

```
When Play starts
Repeat 4 times:
    Play costumes: 2 (medium)
```

Why this works

Costumes are different looks of one sprite. The Play costumes block shows them in sequence. Two flipper positions repeated four times form a simple animation. This is different from moving the whole sprite across the stage.

Try, watch, fix

Play and look only at the body. If it jumps, align the drawings in the Paint Editor. If costumes are missing, check the app version and that both costumes are included.

Make it yours

Add a third costume between flippers open and curled. Compare the movement. Or choose an emoji with multiple costumes instead of drawing.

Tell someone

Show two still drawings before you show the animation. Let your partner explain how still pictures become motion.

The day the lights changed

Backdrops and scene-start events

At lunchtime the children's painted shore was full of bright umbrellas and footprints. In their evening picture the umbrellas were packed away, but a wrapper still lay on the sand. Thandi added a paper bin. Sipho moved the drawn wrapper into it before they finished the second picture. Their octopus narrator watched from its imagined sea. Gogo said litter left on the ground could end up in the ocean. Thandi had thought a sunset alone would end the story. Now she wanted the ending to show care as well as colour. She placed the pictures in order: first the busy afternoon, then the quiet shore with nothing left behind by their visitors.

Keep the story with you

How will your reader know that time has changed?

Make the scene

1. Add two library backdrops and name them in your notes as day and evening. Use any two clearly different scenes; they do not need literal names in the app.

2. Add an octopus narrator sprite. Its Play stack switches to your first backdrop instantly, waits 3 seconds, then switches to the second.

3. On the same narrator, add a separate backdrop-start stack for the evening backdrop, followed by a speech bubble.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

NARRATOR: TIMELINE

When Play starts

Switch backdrop to first scene (instant)

Wait 3 seconds

Switch backdrop to evening scene

NARRATOR: EVENING RESPONSE

When evening backdrop starts

Say in bubble: Take your litter home or use a bin. for 3 seconds

Why this works

A backdrop can start another stack. The official guide says this event begins halfway through the transition. Choose instant transitions if your scene needs a clear immediate change, and test the timing.

Try, watch, fix

Play twice. The first scene should return before the evening change. If the response never comes, make sure the event and switch block select the same second backdrop.

Make it yours

Add one sound for each scene. Record your own quiet evening narration with permission, or keep the story completely visual.

Tell someone

Tell a story with two paper pictures first. Then explain which event moves your digital story to its next moment.

One story, two voices

Internal messages and sound sequencing

At the sea-care puppet show, Sipho and Lerato both began speaking at once. Their a fish puppet and a turtle puppet seemed to argue, and the little audience heard a tangle of words. Sipho stopped. Lerato stopped too. Someone giggled in the silence. Gogo put a smooth craft stone between them. Whoever held it told the next part. Sipho's fish welcomed the audience. Then he passed the stone, and Lerato's turtle asked them to use a bin. The stone did not contain their words. It only told them whose turn had arrived. By the end of the show, everyone had followed the story. Sipho left the stone beside the puppets for next time. Some tools helped people listen.

Keep the story with you

How can code pass a turn without making two voices overlap?

Make the scene

1. Add two storyteller sprites: a fish and a turtle. Add a short library sound for each, or record two brief original lines with permission.

2. On storyteller A, play the first sound 'until done', then Send a message using the message menu. Choose an existing message and note its exact choice.

3. On storyteller B, add When message received with the identical choice, then play the second sound 'until done'. This is an in-project message, not a message sent to a person.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

STORYTELLER A

When Play starts

Play first line (until done)

Send chosen message to all

STORYTELLER B

When chosen message is received

Play second line (until done)

Why this works

Send hands over a signal. The receive block is a start block. Only a matching message starts B's stack. The first audio finishes before the signal is sent, so the planned voices take turns.

Try, watch, fix

Play once and listen. A should finish before B begins. If B stays silent, compare the message choices. If B starts immediately, check that its stack begins with Receive, not Play.

Make it yours

Let B send another message for a third storyteller. Use captions too, so the meaning is available without hearing.

Tell someone

Explain the difference between this local code signal and exporting a project file to someone else.

A glow in our sea story

On-screen glow and colour

The children set their paper sea on a dark blue cloth for the puppet show. Lerato wanted their jellyfish to stand out without lighting the whole room. She tried a yellow outline, then a blue one. Sipho thought yellow felt cheerful. Thandi thought blue felt quiet. Neither colour was the correct answer. Each changed how their imaginary underwater scene felt. On the screen they gave their jellyfish a soft border of light. Gogo asked if all real jellyfish glowed like that. "No, this is our story effect," Lerato said. They were using colour to guide the reader's eye, not pretending their artwork was a photograph of nature.

Keep the story with you

What mood can you create with colour rather than words?

Make the scene

1. Paint a jellyfish sprite on a dark library backdrop. Select it.

2. Attach Glow with a warm colour below the Tap-start block. Then attach Set colour for the sprite, choosing a colour you like.

3. Change the colour options and compare. Do not use the physical flashlight block for this project.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

JELLYFISH

When this sprite is tapped

Glow with your chosen warm colour

Set sprite colour to your chosen colour

Why this works

Glow makes a border of light around the sprite. Set colour changes its colouring. These are on-screen effects. A real torch or flashing light is not needed. To remove a set colour, choose no colour in that block's menu.

Try, watch, fix

Tap once and compare the jellyfish before and after. If its details disappear, change the colour or simplify the drawing. Avoid fast flashes or repeated rapid colour changes.

Make it yours

Make a second scene with a calm cool colour. Ask your partner what feeling it suggests, without telling them your answer first.

Tell someone

Offer a non-glowing version for anyone uncomfortable with light effects. No child needs to watch a flashing screen to learn colour.

The sea-care dance

Shake event and accessible alternatives

The first raindrop landed on Sipho's wrist while he carried the puppet box inside. Another hit the dusty path. Thandi saved the paper crab just before it got wet. Indoors, Lerato gave it a small dance to celebrate being safe. Sipho wanted the screen version to begin with a shake. Lerato wanted to tap. Their story had room for both. Gogo pointed to a wrapper near the doorway and put it in the bin rather than let it wash down the street. The children added one line to their sea story: "Litter in the bin!" The crab could celebrate a careful choice, and nobody had to move in the same way to join in.

Keep the story with you

How can the same project welcome different ways of moving?

Make the scene

1. Choose a crab dancer sprite. Make a Shake-start stack with Jump 20 and a short bubble. Choose low shake sensitivity to begin.

2. Add a second stack on the same sprite starting when it is tapped. Give it the same jump and bubble so shaking is optional.

3. Hold a phone or tablet with both hands over a soft surface. Make a gentle movement only. On Chromebooks, use the on-screen shake button.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

CRAB DANCER: SHAKE ROUTE

When device is gently shaken

Jump 20

Say in bubble: Litter in the bin! for 2 seconds

CRAB DANCER: TAP ROUTE

When this sprite is tapped

Jump 20

Say in bubble: Litter in the bin! for 2 seconds

Why this works

A sensor event connects a physical action to code. A tap event can give the same result. Hardware and movement needs differ, so the accessible alternative is part of the design, not a lesser project.

Try, watch, fix

Try one gentle shake, then one tap. Each should cause one visible response. If shake repeats too easily, adjust its sensitivity or use the tap route only.

Make it yours

Use two costumes instead of a jump if you have them. Keep the movement slow enough to follow.

Tell someone

Ask your partner which route they prefer. Never require them to shake or hold a device in a way that is uncomfortable.

The turtle finds home

Tilt in a loop and sprite collision

Lerato made a path on cardboard using folded strips for walls. Instead of a marble, she slid their flat paper turtle towards a wide blue circle. The circle was its imaginary sea home. Sipho wanted to finish first, but Lerato wanted to steer without bumping every wall. On the next try he slowed down. The turtle arrived, and he looked more pleased than when he had been racing. Later they made a screen version that moved when the device tilted. They left plenty of open space around the home. Their story was about guiding a drawing, not moving a real animal. Its little turtle still needed the same skill: watch, make a small change, and see where it takes you.

Keep the story with you

What happens when you steer with small movements?

Make the scene

1. Paint a turtle sprite and a large painted sea-home goal sprite. Place the goal far enough from the starting turtle that they do not touch. Make a Play stack on the turtle that goes to a chosen start position.

2. After that reset, use Tilt to move from More Blocks inside Forever. It normally comes with a Forever block. Set all four directions and a slow speed. Hold the device at your neutral angle before Play.

3. On the turtle, add a touching-start event targeting the goal. Add a celebration bubble then Stop all scripts. On Chromebook use its on-screen tilt controller.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

TURTLE: STEERING

```
When Play starts
Go to chosen start position (instant)
Forever:
    Tilt to move (four directions, slow)
```

TURTLE: ARRIVAL

```
When turtle touches goal
Say in bubble: Home! for 2 seconds
Stop all scripts
```

Why this works

Forever keeps checking the tilt while the game runs. The touching event needs two sprites. Stop all ends the project's running scripts; the Stop button can also stop it at any time.

Try, watch, fix

Keep both sprites away from edges. Steer gently into the goal. Check that the game stops. Play again to reset the turtle. If a device has no usable tilt, use Chromebook controls or the alternative below.

Make it yours

Alternative: replace the Forever/tilt section with four arrow sprites. On the turtle create four tap-target stacks, each moving it 10 units in that arrow's direction.

Tell someone

Draw the path before playing. Make a wide route rather than a difficult maze. Do not tilt a Chromebook physically.

Three scraps off the shore

Variables, reset and one-time collection

Thandi drew three scraps of litter on a paper beach. Beside them she placed their purple octopus, waiting in its sea. Sipho counted the scraps, then counted again when he lost his place. Gogo gave him three little marks to cross off, one for each scrap moved into a drawn bin. This time he could see what had already been done. When the last scrap was gone, he had three marks and a clear paper shore. They began the next game with the scraps back in place and the count at zero. No real beach rubbish came onto the craft table. A number could remember their pretend clean-up, but only if they told it when a new game began.

Keep the story with you

How can a number remember an action without counting it twice?

Make the scene

1. Paint three visible scrap-of-litter sprites on a shore backdrop. Use drawings, not real beach rubbish. In More Blocks, open a variable menu and use + to create 'scraps'. Variable names cannot currently be deleted, so choose carefully.

2. On scrap A, build Play > Set scraps to 0 > Show at 100%. On scraps B and C use Play > Show at 100%. Variables display by default; use Display variable on if needed.

3. On each scrap, add Tap this sprite > Change scraps by 1 > Hide instant. Once hidden, it cannot be tapped again. Press Play and wait for setup before collecting.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

SCRAP A: SETUP

When Play starts

Set variable scraps to 0

Show at 100% (instant)

EACH SCRAP: COLLECT

When this sprite is tapped

Change variable scraps by 1

Hide (instant)

Why this works

A variable is a named number your code can update. Set makes a fresh zero. Change adds one. Hide prevents the same scrap being collected again. B and C need their own Show-at-Play stacks, not another count reset.

Try, watch, fix

After Play, tap each scrap once. Expect 1, 2, 3. Tap the empty place where a scrap was: it should not add again. Restart and expect 0 with all scraps visible. Tap slowly while testing.

Make it yours

Make five scraps and change the goal. Keep one reset owner so it is easy to find. Test fast taps on the actual device before using this as a classroom game.

Tell someone

Tell your partner why 'change scraps by 0' would not reset yesterday's score.

The turtle says thank you

If, comparison and a meaningful goal

Sipho wanted their paper turtle to say "Thank you!" when someone tapped it. Thandi pointed to a drawn wrapper still lying on the beach. The clean-up game was not finished. Sipho added a check to his plan: count the three removed scraps first. If all three were counted, thank the player. Otherwise, ask them to find what remained. It was kinder than simply saying no. It told the reader what to do next. When the third scrap disappeared into the imaginary bin, Lerato tapped the turtle. This time the thank-you meant something. Gogo reminded them that a real shore needed care beyond their three drawings. The small game was a beginning, not a promise that a whole beach was clean.

Keep the story with you

What should your project do when the goal is not reached yet?

Make the scene

1. Duplicate Project 15. Add a turtle sprite, keeping its code separate from the scraps.

2. On the turtle, attach an If block below Tap this sprite. Insert the scraps variable reporter into its first number slot. Select greater-than and put 2 in the other slot.

3. Inside If put a bubble 'Thank you for clearing the litter!' for 2 seconds. Tap the If block's plus to add Else; inside Else put 'Find all three scraps first.' for 2 seconds.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

TURTLE

```
When this sprite is tapped
If scraps > 2:
    Say: Thank you for clearing the litter! for 2 seconds
Else:
    Say: Find all three scraps first. for 2 seconds
```

Why this works

If checks a condition once, when the tap reaches it. With one point per scrap, scraps greater than 2 means all three are collected. Else is the other path. This event-driven check does not need Forever because every new tap makes a fresh check.

Try, watch, fix

Tap the turtle at 0, 1, 2 and 3 scraps. Only the last should give the thank-you message. Restart and check the turtle gives the unfinished message again.

Make it yours

For five scraps, change the condition to scraps > 4. Say why changing only the artwork would leave the old rule wrong.

Tell someone

Make a two-path paper diagram: goal reached / not yet. Give helpful feedback on both paths.

The fish surprise

Random numbers and bounded choices

Lerato painted three differently sized fish on folded cards. Sipho picked one from a cup, put it back and picked again. The same small fish appeared twice. "That cannot be a surprise," he said. Gogo asked whether he had left it in the cup. He had. They tried again. Small, large, small. Every card still belonged to the choices. Lerato decided her screen fish could change size by surprise too, as long as she chose the limits. They were inventing a picture game, not measuring the size of wild animals. Sipho made one last guess. He was wrong, laughed, and held out the cup for Thandi.

Keep the story with you

Can a random choice repeat, and still be random?

Make the scene

1. Add a fish sprite. On a tap event attach Set size. In its number slot, snap the Random number reporter from More Blocks.

2. Set the random range to 40 through 100. Choose instant size changes so the result is easy to compare.

3. Keep the fish near the middle of the stage. Larger size values may need a smaller original drawing.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

FISH

When this sprite is tapped

Set size to random number from 40 to 100 (%)

Why this works

The rounded reporter gives a number to another block. It does not sit alone as an action. The range sets the possible values. Two taps can give the same value; random does not mean always different.

Try, watch, fix

Tap five times and compare sizes. Each should stay between your chosen limits. If nothing changes, check the reporter is snapped inside the size slot rather than lying loose nearby.

Make it yours

Try random 50-60 and compare with 40-100. Which range makes changes easier to notice? Write a prediction first.

Tell someone

Connect a real cup-of-numbers game with a screen number choice. Both need a clear set of possible outcomes.

Lerato's ocean step

Custom blocks and readable code

Sipho kept remembering the dolphins at uShaka Marine World. Lerato made a dolphin puppet and invented a dance for it: a small jump, a half-turn, then a pause with its fins spread wide. At first she described every action each time. After a while she called it "the ocean step". One name held the little sequence. Sipho could ask for two ocean steps without hearing all the instructions again. When Thandi wanted to learn, Lerato unpacked the name into its parts. The name made the dance easier to share. It did not remove the need to know what was inside it. By sunset each child had changed one part and made a step of their own. Their puppet dance belonged to their imagination, not to a lesson on how real dolphins swim.

Keep the story with you

What repeated sequence deserves a name of its own?

Make the scene

1. Add a dolphin dancer sprite. In More Blocks create a custom block called 'ocean step'. Find its definition and snap Jump 20, Turn 180 clockwise, then Wait 1 beneath it.

2. In the dancer's main stack, put Repeat 2 below Play. Put your new ocean-step block inside Repeat.

3. Do not place the definition inside the main stack. It describes what the new block does when called.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

DOLPHIN DANCER: DEFINE

```
Define ocean step
Jump 20
Turn clockwise 180
Wait 1 second
```

DOLPHIN DANCER: USE

```
When Play starts
Repeat 2 times:
    ocean step
```

Why this works

A custom block gives a sequence a name. Calling it runs the actions in its definition. In OctoStudio custom blocks belong to the sprite that created them. They do not automatically appear for every other sprite.

Try, watch, fix

Play: expect two jumps and two half-turns with pauses. Change the jump in the definition and Play again. Both uses should change because they call the same definition.

Make it yours

Make a second named step with different motion. Use the two names in an order that tells a beginning, middle and end.

Tell someone

Let your partner explain the main stack without listing every action. Then let them open the definition to see how it works.

A story to take home

Export formats and safe review

At the end of the afternoon, Thandi's mother wanted to see the little turtle wave. Thandi could show it on the same screen, but she wanted a copy that would travel home safely. Sipho asked whether a recording would let him change the flippers. It would not. Lerato compared it to the puppet show: a video kept what happened, while the puppets and plan let someone make a new show. Gogo checked every picture and sound before any copy left the room. No addresses, private voices or unapproved photographs. Thandi chose a short recording for watching and kept a project-file copy for making. Their sea story had two ways to travel, and she knew what each one carried.

Keep the story with you

Does your reader need to watch, or to open and change the code?

Make the scene

1. Open a copy of Project 9 or another finished short project. Review every image, word and sound with an adult. Remove personal information before export.

2. Open Share > Record a Video. Start a recording and perform the project using the on-screen controls. Stop within 30 seconds. Use the device's save option if available.

3. Try Share > Make a GIF for a short silent loop, up to 15 seconds. Then Share > Send Project File and save a .octostudio copy locally. You do not have to send any of these to another person.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

A SHORT PERFORMANCE TO RECORD

Use your completed turtle or story project

Press Play and show one clear beginning and ending

Keep video under 30 seconds; GIF under 15

Why this works

A video shows one recorded performance and can include sound. A GIF is a short visual loop without sound. A project file preserves the editable, interactive project and needs OctoStudio to open. The recording does not retain live tap or tilt interaction.

Try, watch, fix

Open the saved recording and check the whole performance. Open the project file in OctoStudio and check its blocks. If the share sheet differs, ask the adult to find a local file-saving route rather than posting online.

Make it yours

Prepare a silent GIF and a narrated video of the same idea. Compare which meaning survives without sound.

Tell someone

Back up before uninstalling or clearing app data. Online sharing may use data and requires the adult's audience and privacy check. No child must publish work to finish this book.

Our sea, our future

Combine, debug and explain a complete project

On the last afternoon, the children placed their paper plans along Gogo's table. There were waving turtles and curling octopuses, a taxi to uShaka Marine World and a beach game that remembered what had been cleared. Thandi imagined a seaside welcome sign, inspired by their visit to uShaka Marine World. Sipho imagined music for a sea-care show. Lerato imagined a shore where visitors took their litter away and left animal homes alone. They did not all picture the same future. They listened until their ideas could share one small scene. Their screen story would not clean a real beach by itself. It could help them show their neighbours one careful choice. Thandi drew an octopus to greet the reader. Sipho drew one last wrapper for the reader to put in a bin. This time the ending began with a hello.

Keep the story with you

What change would you like someone to imagine in your community?

Make the scene

1. Plan a marine-care story with three moments on paper: a welcome, something the reader can do, and an ending. Choose original drawings or library sprites, two approved backdrops and one optional short sound.

2. Build the starter below with a Octopus Guide sprite and a Scrap sprite. Choose one internal message for 'ready' and another for 'ending'; they must be different. Keep the Guide visible on both backdrops.

3. Use matching send/receive choices. Add your own words and one feature from an earlier project: costumes, a custom dance, colour or a counted repeat. Keep that feature short enough to test.

Save a project-file copy when your first working version is ready. Keep the original and experiment in a duplicate. Drawings and library media are always welcome alternatives.



Give it instructions

GUIDE: WELCOME

When Play starts

Switch to welcome backdrop (instant)

Say: Tap the scrap to help. for 3 seconds

Send READY message to all

SCRAP: RESET AND SHOW

Stack 1: When Play starts > Hide (instant)

Stack 2: When READY message received > Show (instant)

SCRAP: TAP TO FINISH

When this sprite tapped

Hide (instant)

Send ENDING message to all

GUIDE: ENDING

When ENDING message received

Switch to second backdrop (instant)

Say: Together we can care for our sea. for 3 seconds

Scrap has three separate stacks: Play > Hide; Receive READY > Show; Tap Scrap > Hide > Send ENDING. The Play stack resets it. The messages organise the story across sprites. Use the exact same message choice at each matching send and receive.

Make it work for someone else

Try, watch, fix

Run from the beginning: scrap hidden during welcome, visible after READY, then one tap changes the scene and hides it. Restart and repeat. Check message choices and reset blocks before adding extra features.

Make it yours

Make the goal your own, not only a litter game. Let a friend try without coaching. Watch where they hesitate, ask what confused them, and change one thing.

Tell someone

Keep an editable backup. Show the story, explain one event and one loop or condition, and name the change you made after testing. Choose a safe export only after adult review.

Your maker's reflection

My story is about ___. My reader can ___. The event that begins my project is ___. My code remembers or repeats ___. At first I expected ___. After testing, I changed ___. My next idea is ___.

A partner's kind review

Ask: Which moment stayed with you? Was there a part you could not find or use? Would captions, a larger sprite or a different input help? Give one specific suggestion, not a score for who was fastest.

When devices are different

No child left out of the project

Camera denied? Paint the object. Microphone denied? Use a library sound and a caption. No tilt? Use the arrow-sprite alternative in Project 14. No shake? Use the tap stack in Project 13. On Chromebooks use the on-screen tilt controller and shake or magnet buttons; do not physically shake a laptop.

Magnet, vibration and torch are optional

The official guide has a magnet-start block, but some Android devices do not support it. It also has vibration: only phones with haptic feedback vibrate, while other devices play a buzz sound. The flashlight block requires suitable hardware and may brighten the stage instead. None is needed for this book. Do not improvise with strong magnets, bright flashes or sensitive equipment in a children's session.

Beam is not a private message

Beam sends a Bluetooth signal to nearby OctoStudio devices; it is different from the in-project Send/Receive used in Projects 11 and 20. The guide says phones and tablets can send; Chromebooks can receive but cannot send. Bluetooth and permission settings affect it. A channel helps separate groups but is not a promise of secrecy.

Optional adult demonstration only: on one phone or tablet, start with Tap > Beam a chosen symbol on a chosen channel. On a nearby device use When that beam is received > a speech bubble. Match both the symbol and the channel, enable Bluetooth with permission, and test at close range. Do not include private data. If it fails, return to an in-project message activity.

Hardware behaviour and accessibility must be checked on the devices actually used. No activity depends on Bluetooth, a magnet, a torch or vibration. Offline-first does not mean every device has every sensor.

A facilitator's first run

Before teaching all 20

1. Check actual device models, app versions, language, storage and permissions. Use one Android or iOS device per pair where possible; include Chromebook variations if that is your equipment.

2. Run Projects 1-5 yourself with drawings and library media. Confirm tap-target menus, size resets, loop placement and hide/show behaviour. Capture genuine screenshots only after the chosen language and version are fixed.

3. Test media and costumes in Projects 6-9. Check permission-denied alternatives. Test narrator timing, scene change and matching internal messages in Projects 10-11.

4. Check shake/tilt and collisions in Projects 13-14. Trial the arrow-button alternative with someone who cannot or does not want to tilt. Confirm the arrival stops reliably.

5. Test score collection slowly and quickly in Projects 15-16. Verify one point per litter scrap, restart to zero and correct turtle feedback. If fast taps can double-count on a device, treat that as a bug to fix before teaching.

6. Test the local backup/open route and all three exports in Project 19. Then trial Project 20 from a clean start, including repeated restart, early tapping and changed messages.

Teach in short making cycles

Read the story aloud without interrupting it to name blocks. Ask one meaningful question. Demonstrate one new screen action, then let pairs try it. Give time to change the artwork and words. End with showing, explaining and keeping a copy, not a race to finish. These are projects, not a fixed one-project-per-week timetable.

Before release: check cultural and language fit with local readers, child comprehension, media rights, safe use and print readability. Do not claim DBE approval or a completed CAPS mapping from this manuscript.

See what a learner understands

Listen to explanations, not only results

Use four descriptions: "I tried it with help", "I can do it", "I can explain it", "I can change it for someone else". They are prompts for support, not labels attached to a child. Record only what is needed for the class; keep individual records private.

After 1-5

Find Home and the editor. Select the correct sprite. Match a start event to an action. Predict a repeat. Reset an initial value.

After 6-10

Choose media with permission. Explain the difference between sprite and backdrop. Make two aligned costumes. Explain a scene transition.

After 11-14

Match an internal message. Give an alternative to a sensor input. Use a loop for continuous steering. Test with two sprites.

After 15-18

Reset and change a variable. Explain one condition. Recognise a random choice can repeat. Use and unpack a custom block.

After 19-20

Choose between an editable file and a recording. Remove private content. Explain the project. Describe one improvement made after another person tried it.

STEAM belongs in the work

Science: observe artwork details and compare predictions with outcomes. Technology: understand media and device differences. Engineering: plan, build, test and change an interaction. Art: draw, compose, narrate and design for an audience. Mathematics: count turns, changes, ranges and goals. Marine care gives the projects a shared purpose; art carries their meaning.

Your little dictionary

Backdrop / The scene behind the sprites.

Block / One instruction or value in the app.

Condition / A test with a true or false answer.

Costume / Another look or pose for the same sprite.

Custom block / A named sequence you define for one sprite.

Event / Something that starts a stack: play, tap, a message or another trigger.

Export / Make a copy outside the app, such as a video or project file.

Loop / Code that repeats. Repeat counts; Forever continues until stopped.

Message / A signal between sprites. A beam is a different nearby-device signal.

Reporter / A rounded block that supplies a value inside another block.

Reset / Give the project its starting values again.

Sequence / Instructions in a planned order.

Sprite / A character, picture or object you can code.

Stack / script / Connected blocks that run together.

Stage / The area where the story appears.

Variable / A name that stores a number the project can change.

Three words to keep

Plan. Test. Change. You do not need your first idea to be your last idea.

Sources and credits

OctoStudio was created by the Lifelong Kindergarten group at MIT Media Lab. To learn more, visit octostudio.org. This is an independent Circuit Kids companion book, not an OctoStudio or MIT publication or endorsement.

Stories, teaching text and schematic diagrams in this book were written for Circuit Kids. No official coding-card artwork or screenshots have been reproduced. The code recipes describe documented functions in original project contexts. Original book material is licensed under CC BY-NC-SA 4.0 (see page 2). Production-rights clearance remains a release gate.

Current Reference Guide, July 2026, v1.1.0+
https://resources.octostudio.org/ReferenceGuide.pdf

Reference-guide landing page
https://octostudio.org/support/learning-resources/octostudio-reference-guide

Learning resources
https://octostudio.org/learn/

OctoStudio in Schools
https://octostudio.org/learn/octostudio-in-schools

Bring a Book to Life
https://octostudio.org/learn/bring-a-book-to-life

Age guidance: especially ages 7-12
https://octostudio.org/support/faq/what-age-group-is-octostudio-designed-for

Offline guidance
https://octostudio.org/support/faq/do-i-need-an-internet-connection-and-data-to-use-octostudio

App privacy policy
https://octostudio.org/privacy-app

Primary sources: official documentation, official educational resources and published policy/FAQ. No physical-device trial was possible for this edition. Older search results can surface the July 2025 guide, which omits newer costume features.

Marine story grounding (10 October 2026):
https://www.saambr.org.za/aquarium/
https://www.saambr.org.za/turtles/
https://saambr.org.za/thandi-the-turtle-litter-campaign-2/
https://ushakamarine.com/seaworld/
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Optional resources, not included assets:
https://octostudio.org/support/learning-resources/octostudio-coding-cards
https://resources.octostudio.org/OctoStudio-Coding-Cards.pdf

Learning and print checks

1. This edition is for ages 7-12, with township storytellers and a marine story arc.

2. Projects use two-page A4 spreads. Project 20 has an extra testing/reflection page.

3. Read and make together. Talk about the stories and caring for the sea.

4. Test projects on the devices you will use before teaching a group. Check the expected results and adapt the pace to your learners.

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Free digital edition. Physical print preparation and device trials remain to be completed.